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Roll No

AU-705-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Computer Aided Engineering and FEM

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the isoparametric concept in finite element analysis.
b) Discuss the convergence criteria for isoparametric elements.
2. a) What is structural analysis? State its objectives.
b) Describe the Modeling of infinite d.o.f. system into finite d.o.f. system. Why it is done?
3. With the help of flow chart explain the basic steps used in finite element problem formulation.
4. Derive and explain the axis-symmetric formulation for deformation of elemental volume with neat sketch.
5. State and explain the need of global co-ordinate system in FEA.

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6. For a higher degree of freedom element explain the use of direction cosines [$l = \cos\theta$ $m = \cos(90-\theta)$] in expressing the displacement vector in global coordinate system. Here θ is the inclination of the element from horizontal.
7. Explain the use of quadratic and cubic elements in analysis of Iso-parametric element.
8. Explain the following element with their interpolation function (shape function)
 - a) 2 D rectangular element
 - b) 1 D cubic element
